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Surgical Treatment in Proximal Bile Duct Cancer

A Single-Center Experience

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Objectives

The authors evaluated the experience and results of a single center in surgical treatment of proximal bile duct carcinoma.

Summary Background Data

Whenever feasible, surgery is the appropriate treatment in proximal bile duct carcinoma. To improve survival rates and with special regard to liver transplantation, the extent of surgical radicalness remains an open issue.

Patients and Methods

Retrospective analysis of 249 patients who underwent surgery for proximal bile duct carcinoma via the following procedures: resection (n = 125), liver transplantation (n = 25), and exploratory laparotomy (n = 99). Survival rates were calculated according to the Kaplan-Meier method, uni- and multivariate analysis of prognostic factors, and log rank test ($p < 0.05$).

Results

Survival rates after resection and liver transplantation are correlated with International Union Against Cancer (UICC) tumor stage (resection: overall 5-year, 27.1%; stage I and II, 41.9%; stage IV, 20.7%; liver transplantation: overall 5-year, 17.1%; stage I and II, 37.8%; stage IV, 5.8%). Significant univariate prognostic factors for survival after liver resection were lymph node involvement (N category), tumor stage, tumor-free margins, and vascular invasion; for transplantation, they were local tumor extent, N category, tumor stage, and infiltration of liver parenchyma. For resection and transplantation, a multivariate analysis showed prognostic significance of tumor stage and tumor-free margins.

Conclusion

Resection remains the treatment of choice in proximal bile duct carcinoma. Whenever possible, decisions about resectability should be made during laparotomy. With regard to the observation of long-term survivors, liver transplantation still can be justified in selected patients with stage II carcinoma. It is unknown whether more radical procedures, such as liver transplantation combined with multivisceral resections, will lead to better outcome in advanced stages. With regard to palliation, surgical drainage of the biliary system performed as hepatojejunostomy can be recommended.

Bile duct cancer or cholangiocarcinoma can be differentiated in the intrahepatic peripheral and the extrahepatic type originating from the main bile ducts. The latter includes the proximally located hilar tumor of the bifurcation with a subtype of the so-called "Klatskin" tumor.[1] With regard to extension in the right and left bile duct, this tumor can be classified according to Bismuth and Corlette.[2] The initial symptom frequently is obstructive jaundice. Because of anatomic reasons, surgical treatment is burdened by many technical problems. However, there is agreement in the literature that whenever feasible, surgery is the appropriate option for prolonged survival and potential cure.[3–13] During the past few years, the following issues have been discussed[14–25]:

The value of preoperative biliary drainage to decrease postoperative mortality;
The extent of surgical radicalness, with special regard to hepatic resection including segment I according to prognostic factors;
The value of liver transplantation in unresectable cases—which may be extended by multivisceral resection or even cluster transplantation; and
Palliation by surgical or nonsurgical biliary drainage.

In the following article, we attempt to evaluate these issues from a single-center experience with 249 patients.

PATIENTS AND METHODS

Within a period of 18 years (1975–1993), 249 patients with proximal bile duct carcinoma underwent surgery at our institution. There were 146 men and 103 women with a mean age of 54.2 ± 11.3 years (median age, 54 years; range, 22–82 years). Based on bile duct location, tumors could be classified according to Bismuth: type I, n = 7 (2.8%); type II, n = 14 (5.6%); type III, n = 97 (39%); and type IV, n = 131 (52.6%). One hundred twenty-five patients (50.2%) underwent tumor resection as follows: hilar resection, n = 30; hilar resection combined with partial hepatectomy, n = 95 (including segment I, n = 43). In 33 patients, segmental resection and reconstruction of the portal vein was performed—of the hepatic artery in 1 and of both vessels in 2. Twenty-five patients (10%) had liver transplants; 18 of these patients had been considered to be unresectable; and 7 of the patients had incidental carcinomas while transplantation was performed for the diagnosis of primary sclerosing cholangitis. Because of locally advanced tumors in 99 patients (39.8%), tumor resection or liver transplantation could not be realized; 57 of these patients were surgically explored with bile ducts pre- or postoperatively drained percutaneously, and 42 underwent the following different kinds of surgical drainage procedures—21 combined with tube implantation: intrahepatic cholangioenteric anastomosis as so-called hepatojejunostomy (n = 18), hepaticojejunostomy (n = 18), choledochojejunostomy (n = 6; Table 1).

In cases of resection or liver transplantation, histopathologic examination of the specimen led to classification of the tumor according to the TNM staging system suggested by the International Union Against Cancer (UICC).[26] The distribution of the tumors according to the pTNM categories is depicted in Table 2.

Local tumor excision as hilar resection without or with combined partial hepatectomy of different extent and liver transplantation were performed according to standard techniques and described previously.[27,28]

Statistical analysis of patient survival (including operative mortality) was performed according to the Kaplan-Meier method, calculated from the time of surgery until March 31, 1994 using SAS (SAS Institute, Inc., Carey, NC) and SPSS software (SPSS, Inc, Chicago, IL). At this time, 3 patients were lost for follow-up; 55 patients were alive, 49 without evidence of tumor recurrence. For comparison of mortality rates, chi square test with Yates correction was used; for comparison of survival rates and for univariate analysis of prognostic factors, log rank test was used; p values less than 0.05 were considered significant.

RESULTS

Surgical Mortality

The 30-day mortality rate after liver resection was 10.5%; after hilar excision, it was only 3.3%, and after

Table 1. TYPE OF SURGICAL TREATMENT IN 249 PATIENTS WITH PROXIMAL BILE DUCT CARCINOMA

Exploratory laparotomy [no. (%)]	57 (23)
Surgical drainage [no. (%)]	42 (17)
Resection [no. (%)]	125 (50)
Hilar resection (no.)	30
Hilar resection + partial hepatectomy (including segment I-resection 43) (no.)	95
Right hemihepatectomy	17
Left hemihepatectomy	43
Extended right hemihepatectomy	24
Extented left hemihepatectomy	8
Nonanatomic	3
Liver transplantation [no. (%)]	25 (10)
Total [no. (%)]	249 (100)

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Table 2. DISTRIBUTION OF PROXIMAL BILE DUCT CARCINOMA (n = 249) ACCORDING TO pTNM SYSTEM AND UICC TUMOR STAGE

				Total (n = 150)	Resection (n = 125)	Transplant (n = 25)
pTNM system						
T1	Ductal wall			14	12	2
T2	Perimuscular connective tissue			61	51	10
T3	Adjacent structures			75	62	13
N1	Hepatoduodenal ligament			46	38	8
N2	Other regional					
UICC tumor stage						
Stage	T	N	M			
0	Tis	N0	M0	0		
I	T1	N0	M0	14	13	1
II	T2	N0	M0	45	37	8
III	T1	N1, N2	M0	16	12	4
	T2	N1, N2	M0			
IVA	T3	Any N	M0	70	60	10
IVB	Any T	Any N	M1	5	3	2

combination with partial hepatectomy, it was 12.7%. However, this difference was not significant (p = 0.2664). After transplantation, the 30-day mortality rate was 12%. After hilar resection combined with partial hepatectomy, 10 of 95 patients (10.5%) died of liver failure. Forty-three patients had undergone biliary drainage before tumor resection—26 percutaneous transhepatic drainage, 19 endoscopic stenting, 2 both. Comparing these patients with those without any drainage procedures (n = 74), no significant difference was found for 30-day mortality rate, with 9.3% and 10.9%, respectively, and a postoperative morbidity rate of 52.4% *versus* 65.8% (p = 0.2238). Survival also was not significantly different, with 1-year survival rates of 63.4% *versus* 64.6%, 3-year rates of 20.1% *versus* 37.2%, and 5-year rates of 16.7% *versus* 30.1% (median survival, 18.7 *vs.* 21.1 months; p = 0.2752).

Resection

After liver resection 1-, 3-, and 5-year survival rates were 65%, 33.4%, and 27.1%, respectively (Table 3, Fig. 1). Patient survival was significantly correlated to UICC tumor stage (Table 4, Fig. 2A). No significant difference (p = 0.7000) in survival rate could be found comparing hilar resection and combination with partial hepatectomy (5-year, 28.9% and 26.3%, respectively) (Table 3).

Ninety-one of 125 patients (72.8%) had R0 resection, whereas 27 (21.6%) underwent R1 resection and 5 (4%) underwent R2 resection. Survival was significantly different between R0 and R1/2 resected patients (p = 0.0117; 1-year survival rate, 69.2% *vs.* 52.9%; 3-year survival rate, 40.1% *vs.* 12.2%; and 5-year survival rate, 31.7% *vs.* 12.2% [median survival, 25.7 *vs.* 12.7 months]). Survival also was significantly different between R1 resected patients and those who just underwent exploratory laparotomy (p = 0.0053; 1-year survival rate, 52.9% *vs.* 27.1%; 3-year survival rate, 9.6% *vs.* 3.7%; 5-year survival rate, 9.6% *vs.* 0% [median survival, 12.7 *vs.* 5.6 months]) (Fig. 3).

Liver Transplantation

After liver transplantation 1-, 3-, and 5-year survival rates were 60%, 21.4%, and 17.1%, respectively, and correlated to UICC tumor stage (Table 4, Figs. 1 and 2B). No significant difference was observed with comparison to survival after resection (p = 0.2935; Fig. 1).

Four transplant patients survived beyond 5 years; one patient died of tumor recurrence after 5.3 years, which had led 1 year before to Whipple procedure including right nephrectomy and hemicolectomy. The other three patients still are alive for 8, 9, and 10 years, respectively, without evidence of tumor recurrence. Seven of the transplant patients underwent hepatectomy for primary sclerosing cholangitis. The cancer was an incidental finding during histologic examination. Five of these patients had advanced tumor stages IVA and B, whereas only one patient with a stage II tumor lived for 5 years after transplantation and died of late tumor recurrence.

Table 3. PERCENT SURVIVAL IN PROXIMAL BILE DUCT CARCINOMA (n = 249) ACCORDING TO SURGICAL PROCEDURE

	n	30 days	1 yr	3 yr	5 yr	Median (mo)
Hilar resection	30	96.7	57.8	33.0	28.9	17.9
Combined with partial hepatectomy	95	87.3	67.2	33.4	26.3	19.9
Liver resection	125	89.5	65.0	33.4	27.1	19.9
Liver transplantation	25	88.0	60.0	21.4	17.1	15.5
Surgical drainage	42	84.1	28.1	4.0	0	5.5
Exploratory laparotomy	57	83.0	26.9	2.9	2.9	6.1
Total	249					

Prognostic Factors

An univariate analysis of prognostic factors with regard to resection and liver transplantation revealed the following to be significant. For resection, the N category, tumor stage, vascular infiltration, and R0 resection were significant factors. No significance could be shown for the T category, Bismuth classification, extraductal tumor, perineural and hepatic infiltration, tumor grading, the extent of resection in terms of hilar excision *versus* combination with partial hepatectomy, and additional caudate lobectomy. For liver transplantation, significant factors were T and N categories, tumor stage, and hepatic infiltration. For details, see Table 5. The multivariate stepwise analysis for resected patients revealed significance ($p = 0.0008$) for tumor stage (I and II *vs.* III and IV: relative risk 2.001; 95% confidence interval, 1.237–3.238)) and R0 resection (relative risk, 1.861; 95% confidence interval, 1.135–3.050). For liver transplantation, only tumor stage (I–III *vs.* IV) reached prognostic significance ($p = 0.0022$; relative risk, 4.569; 95% confidence interval, 1.664–12.544).

In ten long-term survivors with a median survival of 8.4 years (range, 6.7–10.3 years), Bismuth type IV and tumor stage IV occurred in three patients. Six patients had moderate or low differentiated tumors; the tumor showed perineural sheath infiltration in six and even local lymphatic tumor dissemination in two. Bismuth type IV tumors did not differentiate in patients survival according to the type of surgical treatment with regard to deep hilar resection, combination with partial hepatectomy, and liver transplantation.

Palliation by Surgical and Nonsurgical Drainage

The comparison between patients with unresectable tumors who underwent surgical exploration for palliation, either surgical (n = 42) or nonsurgical drainage (n = 29), did not show significant difference for 30-day mortality rates (17.4% *vs.* 10.7%; $p = 0.6674$). No significant difference in survival could be observed ($p = 0.6433$; 1-year survival rate, 27.5% *vs.* 45.2%; and 3-year survival rate, 3% *vs.* 7.5%; median survival, 6.3 *vs.* 6.7 months).

DISCUSSION

In any case of obstructive jaundice, proximal bile duct cancer should be suspected. So far, diagnostic imaging does not satisfy accuracy for the decision about unresectability in this tumor entity. Differentiation from inflammatory changes often is not possible. Multiple biliary abscess formation in the liver can lead to the misdiagnosis of metastases and therefore suggest unresectability. Vascular infiltration suggested from angiography also does not preclude resectability. Therefore, in any case, laparotomy in a specialized hepatobiliary unit should be considered. Because of reports and our own observations about the risk of tumor seeding by biopsy,[29,30] we prefer to avoid tumor manipulation before laparotomy and surgical decisions about resectability. In our opinion, it is not very important to operate without histologically proven diagnosis of

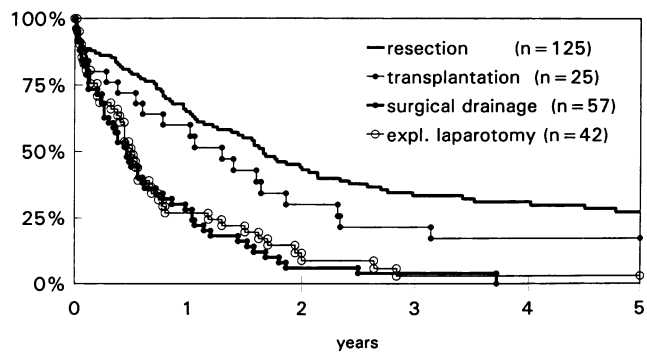


Figure 1. Actuarial survival in proximal bile duct carcinoma (n = 249) according to the type of treatment.

Table 4. PERCENT SURVIVAL IN PROXIMAL BILE DUCT CARCINOMA (n = 249) ACCORDING TO UICC TUMOR STAGE

	n	30 days	1 yr	3 yr	5 yr	Median (mo)
Resection (n = 125)						
I/II	50	94.0	71.6	51.5	41.9	48.3
III	12	91.7	47.6	23.8	0	11.8
IVa,b	63	85.7	62.7	20.7	20.7	16.6
Liver transplantation (n = 25)						
I/II	9	88.9	88.9	55.6	44.4	37.8
III	4	100	75.0	0	0	19.6
IVa,b	12	83.3	33.3	0	0	5.8

malignancy because surgical treatment—even in cases of benign inflammation by sclerosing cholangitis—will just differ in the extent of radicalness.

Value of Preoperative Percutaneous or Endoscopic Biliary Drainage

Liver resection in proximal bile duct cancer is burdened by specific problems due to cholestasis and the risk of liver failure. Because of the reduced ischemic tolerance of the cholestatic liver hilar occlusion, time should be minimalized, and in cases of vascular reconstruction, patency of one of the afferent hepatic vascular systems should be realized. Unfortunately, it is our experience that the "bench procedure," which might be considered appropriate for centrally localized tumors and technical difficulties in vascular reconstruction, is not feasible because the cholestatic liver often will not tolerate this procedure.[31,32] Most patients usually are presented for surgery after implantation of percutaneous or endoscopic biliary drainage catheters. Frequently, preoperative biliary drainage will be performed only for that part of the liver that will be resected, whereas the other portion of the liver that has to be preserved remains cholestatic. This may impair postoperative liver function of the remnant parenchyma and therefore influence morbidity. To decrease the risk of hepatic failure in cases of extended resections, Makuuchi et al.[33] recommend preoperative portal embolization to induce atrophy of the lobe to be resected and hypertrophy of the contralateral lobe to be preserved. In a recent study,[34] this group also showed that the presence of an undrained biliary tree in the right lobe with no biliary stasis in the left lobe induces compensatory hypertrophy of the left lobe and makes extended right hepatectomy a suitable procedure. During the last decade, despite prospective randomized trials, preoperative biliary drainage has been controversial.[14–16,35–37] Our retrospective results do not show a significant advantage with regard to postoperative mortality and survival. Beside the possible risk of tumor seeding along the drainage tube, we also are concerned about cholangitis and therefore attempt to schedule surgery in obstructive jaundice suspicious for proximal bile duct cancer as soon as possible. In some cases of advanced cholestasis, and additionally when the mass of preservable liver parenchyma seems to be very low, surgical drainage of that part of the liver can be performed in a first step by intrahepatic cholangioenteric anastomosis or so-called hepatojejunostomy. After recovery of liver parenchyma, hilar resection will be feasible with lower risk. Furthermore, considering livertransplantation in cases of unresectability, hepatojejunostomy can be used as "bridging"—in our series, performed in three patients—to obtain biliary drainage during the waiting period for a donor organ.

Hilar Resection and Combination with Partial Hepatectomy

Bismuth et al.[9] recommended a formalized type of surgical approach according to their classification: type

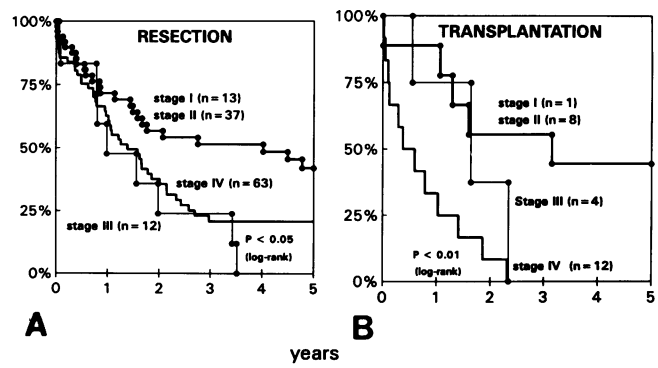


Figure 2. Actuarial survival in proximal bile duct carcinoma after (A) resection and (B) liver transplantation according to International Union Against Cancer (UICC) tumor stage.

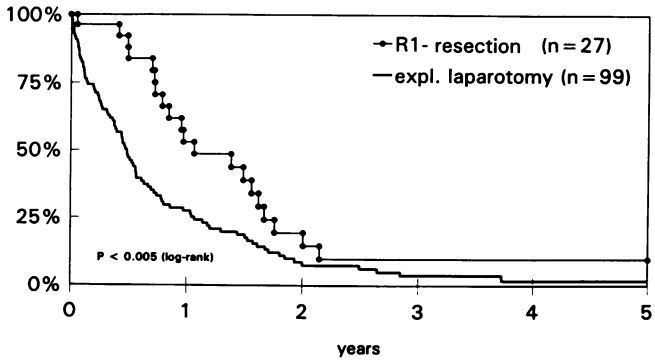


Figure 3. Actuarial survival in proximal bile duct carcinoma after R1 resection and exploratory laparotomy.

I, local excision; type II, local excision and resection of segment I; type III, local excision, resection of segment I, and partial hepatectomy; type IV, hepatectomy and liver transplantation. In our experience, because of sclerosis and fibrosis of surrounding tissue, even surgical exploration cannot always reveal the true extent of the tumor with special regard to bile ducts and liver parenchyma. Therefore, classification of the tumor according to Bismuth and decisions about necessity and extent of hepatectomy often may be extremely difficult. In our series, hilar resection was performed in 30 patients, and according to the intraoperative findings of tumor extension, in combination with partial hepatectomy in 95 patients. Between these types of treatment, no significant difference in 5-year survival rates could be found (28.9% and 26.3%, respectively). The 30-day mortality rate was not significantly lower in hilar resection (3.3% *vs.* 12.7%), although ten patients died of hepatic failure after liver resection.

In some cases, hilar resection was possible—even in Bismuth IV tumors—and the survival rate did not differ from that in patients who underwent combined resection or liver transplantation (Fig. 4). According to the literature[3-13] (Table 6) and our results, 5-year survival rates of approximately 25% can be expected after tumor resection. In our series, (R0) resection for cure was possible in 91 patients (72.8%), whereas microscopic tumor invasion of the resected bile duct stumps (R1) was found in 27 patients. Outcome after R0 resection was significantly more favorable. Evaluation of survival data according to UICC tumor stage shows correlation and significant differences between stage I/II and III and IV (5-year survival rate, 41.9% in stage I/II *vs.* 20.7.% in stage IV, respectively). This elucidates the important prognostic role of lymph node involvement (stage III) and hepatic parenchyma infiltration (stage IV) and may explain why even in Bismuth type IV tumors without infiltration of liver parenchyma satisfying results by deep hilar resection may be obtained.

For anatomic reasons, with biliary drainage of the caudate lobe into both hepatic ducts, Mizumoto and Suzuki[38] recommended routine resection of segment I for complete removal of proximal bile duct carcinoma. This is supported by the work of Nimura et al.[15] In a series of 66 patients, curative resection was possible in 46. In 45 of these patients, caudate lobectomy was performed (5-year survival rate, 37.8%), and in 44, tumor was microscopically shown in segment I. The technical problems of this procedure were described by Elias et al.[39] According to these authors, resection of segment I associated with right or left hepatectomy can be considered a standard hepatic resection, whereas isolated resection of segment I remains a technical challenge. Although performed with favorable results by routine in most cases of recent Japanese series,[12,17] Bismuth et al.[9] and Baer et al.[11] promoted an approach based on the individual findings. With special regard to Bismuth type I tumors that occurred in our series, in only 2.8% local tumor excision by hilar resection is considered to be sufficient by Bismuth et al.[9] In our cases, resection of segment I also has not always been a standard procedure, whereas in the past few years, it has been performed more frequently in combination with partial hepatectomy. Our results have not shown yet a significant prognostic influence on survival.

Liver Transplantation

Because of disappointing results and with regard to the deficiency of donor organs, the indication for liver transplantation in hepatobiliary malignancy has been controversial during the past few years.[20,40-45] With special regard to the indication for liver transplantation and because of different prognoses, intrahepatic cholangiocellular and proximal extrahepatic bile duct carcinoma

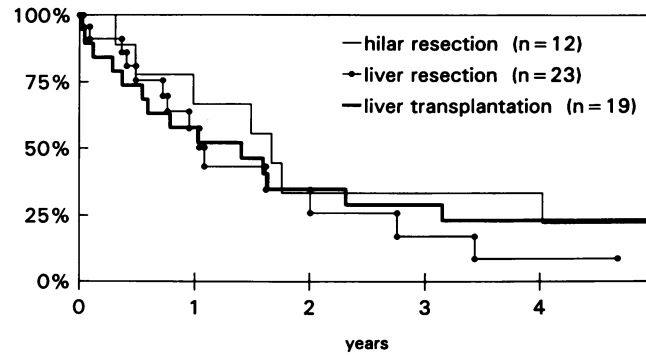


Figure 4. Actuarial survival in Bismuth 4 proximal bile duct carcinoma according to type of surgical treatment.

Table 5. PROGNOSTIC FACTORS IN PROXIMAL BILE DUCT CARCINOMA ACCORDING TO UNIVARIATE AND MULTIVARIATE ANALYSIS IN 249 PATIENTS

	Resection	Transplantation
T	0.0732	0.0117*
N	0.0165*	0.0424*
Tumor stage	0.0184*	0.0049†
R0, R1,2	0.0117*	0.5420
Bismuth classification	0.3961	0.2226
Vascular invasion	0.0092*	0.2814
Extraductal	0.1605	0.6385
Perineural invasion	0.8981	0.4541
Infiltration of liver parenchyma	0.0423*	0.0446*
Grading	0.3641	0.7461
Hilar *vs.* partial liver resection	0.7000	—
Resection of segment I	0.1742	—

* $p < 0.05$ (log rank test).
† $p < 0.01$ (log rank test).

have to be differentiated strictly.[46,47] For proximal bile duct carcinoma, comparison of our results between resection and transplantation reveals no significant difference in outcome (1- and 5-year survival rates of 65% and 27.1% *vs.* 60% and 17.1%, respectively) especially in early tumor stages (94% and 41.9% *vs.* 88.9% and 44.4%, respectively). It must be emphasized that transplantation was performed only in cases of unresectability proven by surgical exploration. This might be interpreted as an advantage for the treatment by transplantation, presuming an identical tumor extension. In our series, 4 of 25 transplanted patients survived more than 5 years—3 still are alive—after 8, 9, and 10 years without evidence of disease. One of our long-term survivors, alive for 10 years, was transplanted successfully in May of 1985 after previous R1 hilar resection of a lymph node-negative stage II tumor performed in our own institution in October of 1984. Seven of the patients included in the analysis underwent liver transplantation for primary sclerosing cholangitis. Carcinoma was discovered incidentally during histologic examination of the hepatectomy specimen—in four patients, an advanced tumor stage IVA and IVB. Only one of these patients with a stage II tumor survived more than 5 years and died of late recurrence.

Currently, we exclude patients with tumor-positive lymph nodes from transplantation because the longest survival in patients with positive lymph nodes was only 16 months. Split-liver transplantation is an alternative that can make more efficient use of the limited resources of donor organs.[32] From our experience, liver transplantation still may be discussed as a therapeutic option in cases of 1) unresectability in presumed UICC stage II confirmed by laparotomy; 2) status postresection with the intention for RO with R1- or R2-positive resection margins due to advanced central tumor infiltration; and 3) local intrahepatic recurrence.

It remains an open issue whether liver transplantation and additional multivisceral resection of the celiac trunk organs and radical lymphadenectomy will improve the results, maybe even in cases of lymph node involvement. After approaches to increase radicalness by *en bloc* resection of the hepatoduodenal ligament,[48,49] Neuhaus and Blumhardt[21] combined liver transplantation and partial pancreatoduodenectomy. Starzl et al.[22] introduced "cluster transplantation." According to the 5-year experience with "cluster transplantation" in different types of primary and secondary hepatobiliary malignancy, Alessiani et al.[50] observed long-term survival (52, 54, and 59 months) in 3 of 12 patients with metastases confined to the liver from extrahepatic bile duct cancer and concluded that this group of patients might benefit from the procedure. In comparison with hepatocellular or intrahepatic bile duct carcinoma, more favorable results of extrahepatic bile duct cancer also must be contributed to the specific biologic behavior and slow tumor growth. All these aggressive procedures seem to be attractive options. Pancreatic transplantation for the indication of advanced bile duct malignancy performed as part of the "cluster" means a potential source of complications. Therefore, in three patients with liver transplantation and multivisceral resection, we performed total pancreatectomy, accepting the problem of diabetes. With regard to the possible benefit of prolonged survival, postoperative morbidity and mortality of these radical procedures have to be evaluated in a larger number of patients.

Prognostic Factors

Ouchi et al.[51] analyzed prognostic factors in 36 patients resected for proximal bile duct carcinoma. Because R1 resected patients did not differ in survival from those resected completely, these authors found the major aim of surgical treatment the reliable relief of the biliary obstruction. This also might be considered an argument for a nonsurgical approach by endoscopic stenting. In our series, there was a significant difference in survival between R1 and R0 resection, although long-term survivors with positive resection margins also were observed. Related to exploratory laparotomy, survival was significantly better after R1 resection, which demonstrates good surgical palliation. Ogura et al.[52] found a significantly higher local recurrence rate (75. *vs.* 28.5%) in patients with a cancer in-

Table 6. RESECTION FOR PROXIMAL BILE DUCT CARCINOMA: REVIEW OF THE LITERATURE

Author	Year	No. of Patients	Perioperative Mortality (%)	Median Survival (mo)	5-year Survival (%)
Boerma*	1990	499	12	21	13
Hadijs	1990	27	7.4	25	22
Tsuzuki	1990	25	4	26	19
Yeo	1990	53	4		11
Nimura	1990	55	6.4		37.8
Reding	1991	82	15.5	24	
Teleky	1991	21	9.5	34	
Bismuth	1992	23	0	24	
Koyama	1993	20	0		25
Baer	1993	23	4.3	23	
Tashiro	1993	34	2.9		25
Washburn	1995	18	11.1		

* Review of the literature 1980–1989.

filtration within 5 mm of the dissected margin in the transverse direction. The prognostic significance of perineural sheath infiltration in the hilar tissue has been emphasized by Bhuiya et al.,[53,54] who found an influence on patient survival (5-year survival rate, 67% *vs.* 32%, respectively). In our series, this could not be confirmed statistically. Even in long-term survivors with a median survival of 8.4 years, six of ten patients showed perineural invasion at the time of surgery. In our univariate analysis of prognostic factors for resected patients, only lymphnode metastases, UICC tumor stage, tumor-free margins, and vascular infiltration had significance. After transplantation, in addition to the T and N categories, tumor stage and infiltration of hepatic parenchyma were significant. The latter observation probably results from an advanced tumor stage, suggesting extrahepatic micrometastases as the promoting factor for tumor recurrence. For the multivariate analysis, significant factors again were tumor stage and tumor-free resection margin. No influence on survival could be found for tumor grading, partial hepatectomy in addition to hilar resection, and resection of segment I. Our data confirm the value of the UICC staging system by strong correlation between tumor stage and survival. Significant influence of perineural infiltration in stage II (one third of our patients) has not been found. However, it might be a suggestion for pathologists to consider further subgrouping, especially in stage II, according to the extent of extraductal tumor spread and maybe in combination with Bismuth's classification.

Adjuvant and Multimodality Treatment

Experience with adjuvant and multimodality treatment in randomized prospective trials is limited. In our patients, a standardized regimen was not applied. There are several reports in the literature with regard to external, internal, and intraoperative radiotherapy partially combined with systemic chemotherapy.[55–61] A recent study by Pitt et al.[61] showed no effect of radiotherapy on survival in localized resectable tumors without metastatic disease. In all these reports, there are no clear data showing a real benefit of radiotherapy for survival of patients resected with the intention of cure, although this may be possible in those with positive resection margins and unresectable tumors.

Surgical or Endoscopic Drainage for Palliation

In unresectable hilar carcinoma, the main intention of palliation is biliary drainage with relief of jaundice-induced pruritus and anorexia, as well as reduction of cholangitis and hepatic failure to improve quality of life and prolong survival. A review of the literature[23–25,62–64] with regard to percutaneous and endoscopic stenting *versus* surgical drainage does not reveal significant differences in median survival. In one recent report by Washburn et al.,[13] the authors found significantly more favorable survival for nonoperative palliation treatment, leading to the recommendation of percutaneous wire mesh stent or endoscopic endoprosthesis. Lai et al.[24] found a significantly lower 30-day mortality in surgically drained patients. Nordback et al.[25] advocate an advantage of surgical palliation with operative placement of large-bore transhepatic stents into a Roux-en-Y limb of jejunum. Our data also do not show a significant difference in 30-day mortality and survival between sur-

gical and percutaneous biliary drainage. However, the significant difference in survival between R1 resected patients and those who only underwent exploratory laparotomy and drainage procedure emphasizes the value of R1 resection as a very efficient palliation. Fortner et al.[65] observed equal survival in patients with microscopically positive margins compared with those who underwent complete resection. This can be explained by slowness of tumor growth, and it stimulates an aggressive surgical approach to achieve at least R1 resection. For surgical drainage, we prefer the aforementioned hepatojejunostomy whenever the surgical risk seems to be calculable. This technique is adequate to the intrahepatic cholangioenteric anastomosis the so-called "round ligament approach" suggested by Bismuth and Corlette[2] and Guthrie et al.[66] Patients are not bothered by percutaneous drainage tubes and cholangitis is less frequent than in endoscopic intraluminal or percutaneous stenting, as observed by Guthrie et al.[66] (55% *vs.* 19%). However, the procedure requires some technical experience because hemostasis and sewing in the excision area may not lead to luminal narrowing of the opened bile ducts.

CONCLUSION

Because of accompanying sclerosis and fibrosis, it is the typical dilemma in Klatskin tumor that by diagnostic imaging and even by intraoperative exploration, tumor extension may be misjudged. Therefore, the decision about the appropriate surgical approach will remain a crucial issue and a challenge even for an experienced hepatobiliary surgeon. Our retrospective data reveal that even in advanced Bismuth type IV and stage IV tumors, long-term survival can be obtained. Even limited surgery with hilar resection can be curative and lead to long-term survival. R1 resection also may be beneficial for palliation. These are good reasons to justify an aggressive approach. Therefore, the decision about resectability and the appropriate procedure should be made by surgical exploration. During the past few years, the tendency toward more radicalness in locally advanced tumors has led to an increase of resectability and proven technical feasibility with calculable morbidity and mortality. It is an open issue whether further improvement by even more aggressive procedures, including the combination of liver transplantation and multivisceral resection as part of a multimodality treatment regimen, will succeed in further improvement of survival rates. For all further studies, it will be mandatory to stage tumors in a general classification system, *e.g.*, on the basis of the UICC rules.

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